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TRANSMISSION ENGINEERING STANDARD

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1.0 SCOPE

This Transmission Engineering Standard (TES) covers acceptable paint systems, grades of surface preparations, storage, application and safe working conditions in handling primers, top coats and accessories to be used in painting outdoor and indoor structures, buildings and others, intended to be used in the system of the National Grid Saudi Arabia.

2.0 CROSS REFERENCES

This Transmission Engineering Standard shall always be read in conjunction with latest revision of General Specification No. 01-TMSS-01, titled "General Requirements for All Equipment/Materials", which shall be considered as an integral part of this TES.

3.0 APPLICABLE CODES AND STANDARDS

The latest revision/amendments of the following Codes and Standards shall be applicable for the equipment/material covered in this TES. In case of conflict, the vendor/manufacturer may propose equipment/material conforming to one group of Industry Codes and Standards quoted hereunder without jeopardizing the requirements of this TES.

1	78-TMSS-01	Catalytically Cured Epoxy-Phenolic Coating for Atmospheric Service
2	78-TMSS-02	Catalytically Cured Epoxies and Epoxy Phenolics for Immersion Service & Damp Services
3	78-TMSS-03	Primers: Zinc Phosphate
4	78-TMSS-04	Paints: Enamel, Emulsion, Varnish & Lacquers
5	78-TMSS-05	Chlorinated Rubber Systems
6	78-TMSS-06	Aluminum & Steel
7	78-TMSS-07	Non-Skid Deck Paint
8	78-TMSS-08	Miscellaneous Materials for Paints and Repairs
9	78-TMSS-09	Primers- Others
10	78-TMSS-10	Paints for Interiors
11	78-TMSS-11	Coatings and Waterproofing.
12	78-TMSS-12	Floor Systems for Substations

13	78-TMSS-13	Corrosion Protective Paint For Hot Dip Galvanized Steel Structures
14	TES-H-107.02	Color Codes and Standards
15	PME Standards	Presidency of Metrology and Environment Standards (with Environment Protection Development Division)
16.	BS 3900-0:2010	Methods of Test For Paints.

4.0 PAINT AND COATING SPECIFICATION INDEX

4.1 Purpose

The Paint & Coating Specification Index presents in a concise form the recommended paint or coating systems to be applied to specific structures or for specific service conditions. Only items of major importance and conditions with unusual requirements are covered.

4.2 Contents

Approved paints and coatings specifications for the following categories:

Tanks
Vessels
Plant Structures
Coastal Structures
Buildings and Building Materials
Miscellaneous

4.3 Approved Paints And Coatings For Specific Structures

Item	Categories	Section	Surface Prep. Specs	Min. Dft (Mm)	Area To Be Coated / Remarks
4.3.1	Tanks				
a.	Exterior, All Except Chemical	5.5	6.4.1	0.130	Use Aluminum (Section 5.5) for body and piping. Use Black paint (Section 5.6) for stairs, platforms, and handrails and for asphalt tank bodies. Use epoxy (Section 5.1) for wind girders of tanks in coastal areas
		5.6	6.4.1	0.160	
		5.1	6.2.6	0.260	
b.	Oil (Dark Products) Interior Bottoms	5.4	6.4.3.b	0.410	Coat all areas in contact with standing water, including all internals except heating coils. Normally, these areas are as follows: Cylindrical Tanks-bottom and first 600 mm up walls (1.0 m for Section 5.20) Spheroids- Up to lower level of Man way.
		5.20	6.4.4.c	See 5.20	
c.	Oil (Light Products)	5.1	6.4.4.c	0.260	
		5.20	6.4.4.c	See 5.20	
d.	Water, Interior	5.3	6.4.4.c	0.250	Entire interior including all internals

Item	Categories	Section	Surface Prep. Specs	Min. Dft (mm)	Area To Be Coated / Remarks
4.3.2	Vessels				
a.	Exterior	5.5	6.4.1	0.130	Also for attached piping
b.	Horizontal Oil/Brine-containing (interior Bottom)	5.3	6.4.4.b	0.250	Coat entire interior surface including all internals except heating coils. Maximum service Temperature for Section 5.3 is 93° C. Maximum service temperature for Section 5.4 is 70°C
		5.4	6.4.4.b	0.410	
4.3.3	Plant Structures				
a.	Structural Steel and piping, mild exposure	5.5	6.4.4.a	0.130	Paint Stairs, platforms, handrails, with black paint (Section 5.6) Paint the bottom 2.0 meters with black and yellow stripes(Sections 5.6, 5.7)
		5.6	6.4.4.a	0.160	
		5.7	6.4.4.a	0.130	
b.	Structural Steel and piping, immersion or high humidity conditions.	5.5	6.4.1.b	0.130	Paint body of Structures with Aluminum (Section 5.5). Do not paint galvanized steel. Use epoxy system (Section 5.1) where unusually severe corrosive atmosphere exists and for structural steel within 30 meters of shoreline at high tide.
		5.1	6.4.4.a	0.260	
		5.10	6.4.2.b	0.130	
c.	Steel flooring & Grating-Non-skid	5.13	See remarks	Not Critical	Applied over chlorinated rubber under coat (Section 5.10) where non-skid surface is desired.
d.	Structural Steel & piping, High Temperature	5.12	6.4.6.a	0.040 to 0.060	Apply on steel where service temperature is between 150°C to 540°C. Maximum allowable coating thickness is 0.060 mm.
4.3.4	Coastal Structures				
a.	Steel, atmospheric zone	5.1	6.4.4.a	0.260	Coat all exposed steel including galvanized steel. Used for steel above water or on land within 30 m of shoreline at high tide.
b.	Equipment: Electrical and mechanical	5.1	6.4.4.a	0.260	For mechanical and electrical equipment, offshore or within 30 m of shoreline at high tide. Section 5.1 is preferred but often, impractical due to incompatible vendor applied primer or inability to sandblast without damaging the subject or adjacent equipment. Section 5.7 should be used in these cases. Section 5.2 should be factory applied on mechanical or electrical equipment.
		5.7	6.4.1.b	0.130	
		5.2	6.2.7	0.130	

Item	Categories	Section	Surface Prep. Specs	Min. Dft Mm	Area To Be Coated / Remarks
4.3.5	Buildings And Building Materials				
a.	Steel: Interior	5.7	6.4.1.a	See Remarks	Section 5.15 is preferred system for decorative use. Coating thickness is not critical as long as desired appearance is obtained.
		5.15	6.2.5		
b.	Concrete tile, Plaster	5.8	6.2.8	Not critical	Section 5.8 is an Enamel system. Section 5.9 is latex emulsion. Both systems are for mild services only
		5.9			
c.	Wood	5.8	6.2.8	Not Critical	Section 5.14 is spar varnish for exterior use. 5.15 is a nitro cellulose lacquer for interior use on wood or metal.
		5.9	-do-		
		5.14			
		5.15			
d.	Aluminum and Galvanized	5.1	6.4.4.c	0.260	These are not normally coated. When they are coated, it is imperative that the surface preparation specifications be followed.
		5.5	6.4.1.c	0.130	
		5.6	6.4.1.c	0.130	
		5.7	6.4.1.c	0.130	
e.	Fire Retardant	5.17	6.2.8	Not Critical	Used on interiors of buildings and other enclosures where fire resistance is desirable.
4.3.6	Miscellaneous				
a.	Pavement Markings	5.19	6.2.8	Not critical	Traffic and Zone markings on concrete, asphalt, or Bituminous surfaces.
b.	Fluorescent Enamel	5.16	See Remarks	Variable see 5.16	Applied over white enamel undercoat. For improved visibility on dawn and dusk.
c.	Cables in Substation basement				Apply fire retardant coating paint, rated for 3 hours

Item	Categories	Section	Surface Prep. Specs	Min. Dft Mm	Area To Be Coated / Remarks
c.	Steel Piping & Vessels covered with Insulated Material				
	Cold System (PU Foam)	5.3 only primer	6.2.6	0.130	Apply prior to insulating on steel to be operated intermittently below the atmospheric dew point. No coating needed where operating temperatures exceed 65^0C
	Cold System (Foam Glass)	5.18	6.2.6	0.080	
	Hot System				
	Fire Proofing				Prime bare steel with epoxy or epoxy-Phenolic before applying fireproofing concrete. Steel already shop primed shall have defective areas touched up with compatible coating prior to fireproofing
	Primer/ Topcoat	5.1	6.2.6	0.180	
	Primer only	5.3	6.2.6	0.130	
d.	Shop applied Coatings				Vendor/ Contractor painting shops should use Paint system approved by NG whenever possible. However, other paint systems are acceptable as long as the following conditions are met: 1. They are compatible with end use of the product. 2. They can be maintenance painted with standard paint system and without unusual surface preparation, and, 3. Their color complies with the requirement of TES-H-107.02.
e.	Field Welds,	5.5	6.2.5	0.130	Industrial and residential atmosphere services. For unusual corrosive atmosphere and/or within 30 m of shoreline at high tide.
	Mild Exposure	5.7	6.2.5	0.130	
	Field Welds	5.1	6.2.6	0.260	
	Severe Exposure	5.18	6.2.6	0.060 to 0.10	

5.0 NATIONAL GRID SA PAINT AND COATING SPECIFICATION

5.1 Epoxy Coating System for Atmospheric Service

5.1.1 Type of Coating:

Catalytically cured epoxy coating system, including primers pigmented for corrosion resistance.

5.1.2 Use:

Primarily for coastal, industrial, or other severe atmospheric service up to 150°C, Resistant to many acids, alkalies, and salts (see manufacturer's literature for specific environments). Also suitable for immersion service in tanks and vessels containing light petroleum products such as kerosene and diesel at temperature up to 82°C.

5.1.3 Reference Purchase Specification:

78-TMSS-01

5.1.4 Minimum Acceptable Dry Film Thickness (DFT):

Primer	0.800 mm
Topcoat	0.180 mm
Total	0.980 mm

5.1.5 Approved Material:

Description	Application Method	Min/Max. Recoat Intervals (Hrs) (3)		Curing Time (1)	Material Take-Off Guide (2)	
		25°C	40°C		Typical Coverage m^2/l	Typical Dft Per Coat mm
Primer	Brush or spray	8/72	3/24 4/24		8.6 4.5	0.040 0.080
Topcoats	Brush or spray	8/72	4/24	5/3 Days	8.6 3.9	0.050 0.090

Methyl Ethyl Ketone (MEK)	Equipment Cleaner and thinner (mix with 50% Toluene)
Toluene	Equipment Cleaner and thinner (mix with 50% MEK)

NOTE:

(1) Curing time for temperatures of 25° C/40°C. This is the minimum allowable time interval between final painting and putting into service.

(2) For ordering purposes only. Not a mandatory painting specification.

(3) When recoat intervals exceed the specified maximum times or a new epoxy finish is needed over an existing coating, a softening solvent can be used for atmospheric service conditions. Remove oil, grease and the contaminants from surface, apply solvent, allow to dry 30 minutes, and then apply the epoxy topcoat.

5.1.6 Surface Preparation Requirements:

a. Atmospheric Service: As per Section.6.4.4.a

b. Internal Surfaces/Immersion Service: As per Section.6.4.4.b

c. Minimum/Maximum acceptable anchor pattern (surface profile) is 0.050 to 0.080 mm. Desert sand is not allowed as the blasting medium, only proper size grit will be used for internal surfaces of tanks/vessels. Use angular abrasive grit.

5.1.7 Additional Remarks:

a. Shelf life: 12 months at 40°C.

b. Coating should be thoroughly mixed with power stirrer. After mixing the catalyst into the base portion, the mixed coating should be allowed to stand (sweat in) for 15 to 30 minutes before use. Consult manufacturers' instructions.

c. Rollers will not be used. Preferred method of application is by airless spray.

d. The catalyst is strongly alkaline and should be washed off immediately with water if splashed on the skin.

e. All welds, corners, and edges must be ground smooth and painted prior to application of first spray coat to ensure adequate coverage in these areas.

5.2 Epoxy Polyester Powder Coatings

5.2.1 Type of Coating:

Epoxy-polyester coating, applied to zinc phosphated steel.

5.2.2 Use:

Primarily for atmospheric service below 100°C, resistant to salt water, some diluted acids, oil, petroleum, glycerin and diluted caustic soda and ammonia.

5.2.3 Purchase:

This painting system is recommended for factory application only.

5.2.4 Minimum Acceptable Dry Film Thickness (DFT):

Zinc/Phosphate Primer	0.090 mm
Topcoat	0.040 mm
Total	0.130 mm

5.2.5 Approved Material:

Description	Application Method	Curing Schedule(L)	Typical Dft Per Coat(2) Mm
Primer	Spray or immersion		0.110
Topcoat	Spray or Immersion	200°C -8 min 180°C - 15 min	0.050

NOTE:
(1) Curing time for temperatures of 180°C/200°C. This is the minimum allowable time interval between final painting and putting into service.
(2) Not a mandatory painting specification.

5.2.6 Surface Preparation Requirements:

Mild and Severe Services: paragraph 6.2.7

5.2.7 Material Requirements:

a. Purity of zinc shall not be less than 98%.

b. Coating properties shall be tested as per BS-3900. Test Report shall be provided to National Grid SA.

5.2.8 Additional Remarks:

a. When cured coatings are slightly damaged, which is often caused by jigging, repair can be done with a liquid or air-drying paint specially formulated for use with the paint manufacturer powder.

b. The time interval between shot blasting and zinc application shall not exceed 4 hours. Articles must not be exposed outdoor or to a damp atmosphere prior to zinc application after shot blasting.

c. Application of etch primer shall follow immediately after 2 hours of zinc application.

d. Zinc/Phosphate primer shall be allowed to air-dry before applying the powder coatings.

e. An alternative method wherein zinc, phosphate and warm water are combined to form a solution and applied as one is acceptable.

5.3 Epoxy and Phenolic Epoxy Coatings for Immersion Service

5.3.1 Type of Coating:

Catalytically cured epoxy or Phenolic-epoxy systems.

5.3.2 Use:

Immersion service in water at all temperatures up to 93°C. Typical uses are in storage tanks for raw, softened, treated, distilled, or drinking water and in containers handling hot crude oil/ brine solutions.

5.3.3 Reference Purchase Specifications: 78-TMSS-02

5.3.4 Minimum Acceptable Dry Film Thickness (DFT):

Total 0.250 mm

5.3.5 Approved Material:

Description	Application Method	Min./Max Recoating Intervals (Hours)	Curing Time(3) Days	Material Take-Off Guide(4)	
				Typical Coverage m^2/l	Typical dft Per Coat mm
Primer (1) (Epoxy)	Brush	4/16	5	12.3	0.040
	Spray	4/16	5	4.2	0.100
Top Coat (Epoxy)	Brush	16/48	5	12.3	0.040
	Spray	24/48	5	4.2	0.100
Primer (2) (Phenolic Epoxy)	Brush	2/24	5/2	10.1	0.040
	Spray	12/24	5/2	2.4	0.130
Top Coat(2) (Phenolic Epoxy)	Brush	2/24	5/2	10.1	0.060
	Spray	12/24	5/2	2.4	0.130

MEK Thinner and equipment cleaner (mix with 50 percent toluene).

Toluene Thinner and equipment cleaner (mix with 50% MEK or MIBK).

Methyl Ketone (MIBK) Thinner and equipment cleaner (mix with 50 percent Isobutyl toluene)

NOTES:

(1) Preferred system for potable water tanks.
(2) Preferred system for hot (70 to 93°C) water that is not potable.
(3) Curing times for surfaces at 21°C/380°C. This is the minimum allowable time interval between final painting and putting into service.
(4) For ordering purposes only. Not a mandatory painting specification.

5.3.6 Surface Preparation Requirements: As peer Section 6.4.4

The minimum acceptable anchor pattern (surface profile) is 0.040 mm. Maximum allowable surface profile is 0.060 mm. Desert sand is not allowed as blasting medium, only the proper size approved grit will be used. Use approved abrasive grit for internal surface preparation. Abrasive grit to be angular and free from potential contaminants such as carbonates and chlorides, and recommended by the paint manufacturer as non-contaminating to the prepared surface.

5.3.7 Additional Remarks:

a. Shelf life: 12 months at 40°C.

b. After stirring the catalyst into the resin, the mixed paint may have to stand for 30 minutes prior to application. Consult manufacturers' instructions.

c. Rollers will not be used.

d. The catalyst is strongly alkaline and should be washed off immediately with water if splashed on the skin.

e. Welds, corners, and edges shall be brush painted prior to application of first spray coat to ensure coverage in these areas.

f. This coating system must be certified by a recognized institution for suitability for use in contact with drinking water. Paint Certificate shall bear approval by the Ministry of Health of Saudi Arabia.

g. Continuous ventilation must be provided during curing period to remove all solvent fumes.

5.4 Coal-Tar Epoxy Coatings

5.4.1 Type of Coating

Catalytically cured blend of coal-tar and epoxy resins.

5.4.2 Use:

Primarily for tank and vessel bottoms containing petroleum/brine solutions at temperatures below 70°C. Suitable for immersion in sea water, fresh water, brine, and most alkalies. Resistant to splash and spillage of many acids as well as aliphatic and aromatic hydrocarbon solvents. Temperature limitations: Immersion, 70°C, Atmospheric, 150°C.

5.4.3 Reference Purchase Specification: 78-TMSS-11

5.4.4 Minimum Acceptable Dry Film Thickness (DFT):

Per coat: Min. 0.200 mm, Max. 0.300 mm
Total Min. 0.410 mm, Max. 0.610 mm

5.4.5 Approved Material:

Description	Application Method	Min./ Max Recoating Interval	Curing Time	Material Take-Off Guide (2).	
				Typical Coverage m^2/l	Typical dft Per Coat mm
Primer/ Topcoat	Spray	4/48 hrs	7/3/2 days(l)	3.1	0.200

Toluene Thinner and equipment cleaner. See note 5.4.7.e.

MIBK Thinner and equipment cleaner. See. Note 5.4.7.e.

NOTES:

(1) Curing times for the following conditions; a/b/c: (These are the minimum allowable time intervals between final painting and putting to service.)

a) All temperatures, still air.
b) Temperatures between 21°C to 32°C, moving air.
c) Temperatures above 32°C, moving air.

(2) For ordering purposes only. Not a mandatory painting specification.

5.4.6 Surface Preparation Requirements: As per Section 6.4.3

Minimum acceptable anchor pattern (surface profile) is 0.060 mm.

Maximum allowable anchor pattern (surface profile) is 0.090 mm.

For tank/vessel internal immersion service, desert sand is not allowed as the blasting medium, only proper size approved grit will be used. Use approved abrasive grit for internal surface preparation. Abrasive grit shall be angular and free from potential contaminants, such as carbonates and chlorides, and recommended by the paint manufacturer as non-contaminating to the prepared surface.

5.4.7 Additional Remarks:

a. Shelf life: 6 months at 40°C.

b. Brushes can be used for small areas and are preferred for coating weld beads, fillets, and unusually rough or corroded areas. Rollers will not be used.

c. The catalyst is strongly alkaline and should be washed off immediately with water if splashed on the skin.

d. Whenever possible, welds, corners, and edges should be brush painted prior to application of first spray coat to ensure adequate coverage in these areas.

e. See Vendors instructions to determine whether MIBK or Toluene should be used as a thinner.

5.5 Aluminum Pigmented Coatings

5.5.1 Type of Coating:

Aluminum pigmented alkyd resin.

5.5.2 Use:

Primarily for steel structures, pipelines, and equipment in industrial and coastal atmospheres for temperatures up to 150°C. It is resistant to industrial gases, hydrogen sulphide fumes, high humidity, and spillage of gasoline. Not for immersion or continuous wetting services.

5.5.3 Reference Purchase Specification: 78-TMSS-03 and 78-TMSS-06

5.5.4 Minimum Acceptable Dry Film Thickness (DFT):

Zinc Phosphate Primer	0.080 mm
Aluminum Top Coat	0.050 mm
T o t a 1	0.130 mm

5.5.5 Approved Material:

Description	Application Method	Min./Max Recoating Interval	Material Take-Off Guide *	
			Typical Coverage sq.m/L	Typical Dft Per Coat mm
Zinc Phosphate primer 5/10 Liter cans, 20 liter Drum	Brush or Spray	16 hrs.	7.4	0.040
Aluminum Top Coat		16 hrs	8.6	0.020

Mineral Spirits recommended by the manufacturer to be used as thinner and equipment cleaner.

* For ordering purposes only. Not a mandatory painting specification.

5.5.6 Surface Preparation Requirements: As per Section 6.4.1

5.5.7 Additional Remarks:

a. For surfaces that are cleaned with hand or power, tools, improved coating life can be obtained by applying one coat of a wetting oil pretreatment (1 part boiled linseed oil mixed with 2 parts mineral spirits) prior to applying the zinc Phosphate primer. The wetting oil pretreatment coat should dry to a tacky but not fully dry state before the zinc Phosphate primer is applied. The total dry film thickness requirement is not changed.

b. If paint of a particular batch dries too slowly, up to (but not more than) 0.5 L of drier can be added to each 20 L of paint. Mix well. Excessive addition of drier will ruin the paint.

c. Severe service areas may require power tool cleaning (SP2) or water blasting, where adjacent rotating equipment and/or instruments might be damaged due to normal abrasive blasting techniques.

5.6 Semi-Gloss Black Alkyd System for Metallic Surfaces

5.6.1 Type of Coating:

Graphite pigmented oil-modified alkyd resin.

5.6.2 Use:

Structural steel and equipment in industrial, rural, or coastal atmospheres at temperature up to 120°C where high gloss and gloss retention are not necessary. Resistant to hydrogen sulfide and other industrial fumes and to high humidity. Not for immersion or continuous wetting services.

5.6.3 Reference Purchase Specification: 78-TMSS-03 and 78-TMSS-06.

5.6.4 Minimum Acceptable Dry Film Thickness (DFT):

Zinc Phosphate Primer	0.080 mm
Black Topcoat	0.080 mm
T o t a 1	0.160 mm

5.6.5 Approved Material:

Description	Application Method	Min./Max Recoating Interval (2)	Material Take-Off Guide (1).	
			Typical Coverage sq.m/L	Typical Dft Per Coat Mm
Zinc Phosphate primer 5/10 Liter cans, 20 liter Drum	Brush or Spray	16 hrs	7.4	0.040
Top Coat Black		16 Hrs	7.4	0.040

Mineral Spirits recommended by the manufacturer to be used as thinner and equipment cleaner.

NOTE: (1) For ordering purposes only. Not a mandatory painting specification.

(2) For guidance only. Recoat when dry to touch.

5.6.6 Surface Preparation Requirements: As per section 6.4.1

5.6.7 Additional Remarks:

a For rusty surfaces that are cleaned with hand or power tools, improved coating life can be obtained by applying one coat of a wetting oil pretreatment (1 part boiled linseed oil mixed with 2 parts mineral spirits) prior to applying the zinc Phosphate primer. The wetting oil pretreatment coat should dry to a tacky but not fully dry state before the zinc Phosphate primer is applied. The total dry film thickness requirement is not changed.

b. If paint of a particular batch dries too slowly, up to (but not more than) 0.5 L of drier can be added to each 20 L of paint. Mix well. Excessive additions of drier will ruin the paint.

5.7 High Gloss Enamel System for Metallic Surfaces

5.7.1 Type of Coating

Pigmented oil-modified alkyd resin.

5.7.2 Use

Structural steel, machinery, and equipment in industrial, rural, and coastal atmosphere for temperatures up to 71°C where high gloss and gloss retention are desired. Resistant to hydrogen sulfide and other industrial fumes and to high humidity. Not for immersion or continuous wetting services.

5.7.3 Reference Purchase Specification: 78-TMSS-03 and 78-TMSS-04

5.7.4 Minimum Acceptable Dry Film Thickness (DFT):

Zinc Phosphate Primer	0.080 mm
Topcoat	0.050 mm
Tota1	0.130 mm

5.7.5 Approved Material

Description	Application Method	Min./Max Recoating Interval (2)	Material Take-Off Guide (1).	
			Typical Coverage m^2/L	Typical Dft Per Coat mm
Zinc Phosphate primer 5/10 Liter cans, 20 liter Drum	Brush or Spray	16 hrs	7.4	0.040
High Gloss Enamel		24 Hrs	8.6	0.020

The following materials recommended by the manufacturer shall be used.

Mineral Spirits	Thinner and equipment cleaner.
Enamel Thinner	Solvent for enamels.
Turpentine	Solvent

NOTE:

(1) For ordering purposes only. Not a mandatory painting specification.

(2) For guidance only, Recoat when dry to touch.

5.7.6 Surface Preparation Requirements: As per Section 6.4.1

5.7.7 Additional Remarks

a. Shelf life: 12 months at 40°C.

b. For rusty surfaces that are cleaned with hand or power tools, improved coating life can be obtained by applying one coat of a wetting oil pretreatment (1 part boiled linseed oil mixed with 2 parts mineral spirits) prior to applying the zinc Phosphate primer. The wetting oil pretreatment coat should dry to a tacky but not fully dry state before the zinc Phosphate primer is applied. The total dry film thickness requirement is not changed.

c. If paint of a particular batch dries too slowly, up to (but not more than) 0.5 liter of drier can be added to each 20 liters of paint. Mix well. Excessive additions of drier will ruin the paint.

d. For improved final finish, an intermediate coat of primer-surfacer may be applied after the zinc Phosphate primer. It should be sanded smooth when dry prior to the application of the enamel topcoats.

5.8 High Gloss Enamel System for Non-metallic Surfaces

5.8.1 Type of Coating

Pigmented oil-modified alkyd resin.

5.8.2 Use

Decorative, for interior and exterior non-metallic surfaces in atmospheric exposure for temperature up to 70°C.

5.8.3 Reference Purchase Specifications: 78-TMSS-03 and 78-TMSS-04

5.8.4 Minimum Acceptable Dry Film Thickness (DFT): Not applicable.

5.8.5 Approved Material:

Description	Application Method	Min./Max Recoating Interval (2)	Material Take-Off Guide (1). Typical Coverage m^2/L
Primer-sealer (white)	Brush or Roller	24 hrs	7.4
Undercoater (white)	Brush or Roller or spray	18 hrs	8.6
High Gloss Topcoats		24 hrs	8.6

The following materials recommended by the manufacturer shall be used.

Mineral Spirits	Thinner and equipment cleaners
Enamel Thinner	Solvent.
Turpentine	Solvent.

NOTE: (1) For ordering purposes only. Actual coverage will depend on the porosity of the surface being coated and the applied coating thickness.

(2) For guidance only. Recoat when dry to touch.

5.8.6 Surface Preparation Requirements: As per section 6.2.8

5.8.7 Additional Remarks:

a. For interiors: Apply 1 coat primer-sealer, 1 coat enamel under-coater, and 1 or more coats enamel as necessary.

b. For exteriors: Apply 1 coat primer-sealer and 1 or more coats of enamel as necessary.

c. If a paint of a particular batch dries too slowly, up to (but not more than) 0.5 L of drier can be added to each 20 liters of paint. Mix well. Excessive additions of drier will ruin the paint.

d. A major cause of premature paint failures with this system is moisture build-up under the coating. Plaster and concrete must be completely cured before being painted. If this is not possible, the latex system Section 5.9 should be used in preference to this system.

5.9 Latex Emulsion System

5.9.1 Type of Coating:

Water based latex emulsions.

5.9.2 Use:

Decorative, for interior and exterior non-metallic surfaces in mild, atmospheric service at ambient temperatures. & not recommended for industrial fumes, coastal atmospheres, or frequent wetting.

5.9.3 Reference Purchase Specification: 78-TMSS-04

5.9.4 Minimum Acceptable Dry Film Thickness (DFT): Not applicable.

5.9.5 Approved Material:

Description	Application Method	Typical Recoating Interval (2)	Material Take-Off Guide (1). Typical Coverage sq.m/L
Primer-sealer	Brush or Roller or spray	3-4 hrs	
Topcoats			9.8
Tinting bases for interior use Neutral (5L) Midtore (5L) Midtone (5L) Deep tone (5L)			9.8
Tinting bases for exterior use Midtone (20L)			9.8

Pigments for use with Tinting Bases.

NOTE:

(1) For ordering purposes only. Not a mandatory painting specification.
(2) For guidance only. Recoat when dry to touch.

5.9.6 Surface Preparation Requirements: As per Section 6.2.8.

5.9.7 Additional Remarks:

Finished coating can be cleaned by occasional washing with water and mild soap. This should not be done for at least two weeks after painting.

5.10 Chlorinated Rubber Paint System

5.10.1 Type of Coating:

Synthetic resin made by chlorinating natural rubber and adding inert, acid and alkali resistant pigments.

5.10.2 Use:

Metallic and concrete surfaces at temperatures less than 60°C. Resistant to many acids, alkalies and salts, immersion, high humidity, and mechanical abrasion. Fire retardant. Not resistant to vegetable oils and aromatic or chlorinated solvents.

5.10.3 Reference Purchase Specification: 78-TMSS-05

5.10.4 Minimum Acceptable Dry Film Thickness (DFT):

a. On Steel:

Primer	0.050 mm
Topcoat	0.080 mm
Total	0.130 mm

b. On Non-metallic Surfaces:

Topcoat	0.100 mm

5.10.5 Approved Material:

Description	Application Method	Typical Recoating Interval (3)	Material Take-Off Guide (2).	
			Typical Coverage m^2/L	Typical Dft Per Coat mm
Primer(1)	Brush	24 hrs	7.4	0.020
Topcoat	Brush or spray	24 hrs	7.4	0.040

Enamel Thinner	Solvent
Xylene	Solvent and equipment cleaner.
Toluene	Solvent and equipment cleaner.
MIBK	Used in spraying thinner.
Spray Thinner	1 part MIBK to 3 or 4 parts Xylene or Toluene.

NOTE:

(1) Used only for steel surfaces. There is no special primer for non-metallic surfaces; the topcoats should be applied directly to the cleaned surface.
(2) For ordering purposes only. Not a mandatory painting specification.
(3) For guidance only. Recoat when dry to touch.

5.10.6 Surface Preparation Requirements: See Section 6.4.2

5.10.7 Additional Remarks:

a. Very moisture sensitive during application and drying. Surfaces must be dry.

b. Spraying is difficult and requires an experienced operator to avoid cob-webbing and pin holing. Add no more than 15% by volume spraying thinner.

c. Recommended precautions by the paint manufacturer shall be totally followed.

5.11 Cold-Applied Bitumastic Coatings

5.11.1 Type of Coating:

Solvent cut-back of coal-tar pitch.

5.11.2 Use:

Primarily for corrosion protection of buried or immersed steel at temperatures below 60°C. Resistant to water and acidic fumes. Not recommended for exposure to sunlight. Not resistant to hydrocarbon solvents.

5.11.3 Reference Purchase Specification:

None.

5.11.4 Minimum Acceptable Dry Film Thickness (DFT):

a. Buried:0.760 mm

b. Immersed: 1.140 mm

5.11.5 Approved Material

Description	Application Method	Typical Recoating Interval(2)	Material Take-Off Guide (1).	
			Typical Coverage sq.m/L	Typical Dft Per Coat mm
Coal-Tar base 5/20 liter cans 200 liter drum	Brush or spray	2/3 Days	1.5	0.380

Enamel Thinner Equipment cleaner
Xylene Equipment cleaner

NOTE:

(1) For ordering purposes only. Not a mandatory painting specification.
(2) For guidance only. Recoat when dry to touch.

5.11.6 Surface Preparation Requirements: As per Section 6.4.7

5.11.7 Additional Remarks:

a. Material is thixotropic. Do not add solvents to thin. When stirred vigorously, the bitumastic becomes free flowing. Correct viscosity for application is similar to that of molasses.

b. Use power stirrers. Hand stirring is totally inadequate.

c. Stir from time to time during application as required to maintain the desired viscosity.

5.12 High Temperature Coating System

5.12.1 Type of Coating:

Silicone resins containing zinc dust or aluminum pigments.

5.12.2 Use:

Steel in atmospheric service at temperatures between 150° C and 540° C. Resistance to industrial atmospheres and normal weathering depends on operating temperature.

5.12.3 Reference purchase specification: None

5.12.4 Minimum/ Maximum Acceptable Dry film thickness (DFT):

Total 0.040/0.060 mm.

5.12.5: Approved Materials

Description	Application Method	Typical Recoating Interval(2)	Material Take-Off Guide (1).	
			Typical Coverage sq.m/L	Typical Dft Per Coat mm
Primer (Zinc Rich)	Brush or spray	3 Hours	9.8	0.020
Top Coat (Aluminum color)	Brush or Spray	N/A	9.8	0.020

Toluene Thinner and equipment cleaner.
Xylene Thinner and equipment cleaner.

Note: (1) For ordering purposes only. Not a mandatory painting specification.
(2) For guidance only. Recoat when dry to touch.

5.12.6 Surface Preparation Requirements: As per Section 6.4.6.

5.12.7 Additional Remarks:

a. Probably the least tolerant system in this manual to deviations from the requirements specified above. Any deviation generally results in immediate paint failure.

b. Average life expectancy for temperatures above 400° C is 1 year.

c. The maximum coating thickness of 0.060 mm specified above is critical, because thicker coatings are brittle and spall off during heating up to operating temperature.

d. Most paint failures with this system are attributable to poor surface preparation.

5.13 Non-Skid Floor Paint

5.13.1 Type of Coating:

Phenolic or alkyd-Phenolic resins with sand, alumina, or crushed flint or garnet added for skid resistance.

5.13.2 Use:

A topcoat applied over a fully dried and cured chlorinated rubber paint system, Sections 5.10. For use on such areas as decks, walkways, steps, and floors where slippery conditions are undesirable.

5.13.3 Reference Purchase Specification: 78-TMSS-07

5.13.4 Minimum Acceptable Dry Film Thickness (DFT): Not critical.

5.13.5 Approved Material

Description	Application Method	Typical Recoating Interval(2)	Material Take-Off Guide (1). Typical Coverage Sq.m/L
Non Skid Floor Paint	Brush	18 Hours	6.1

Enamel Thinner Solvent.
Xylene Solvent and equipment cleaner

NOTE:

(1) For ordering purposes only. Not a mandatory painting specification.
(2) For guidance only. Recoat when dry to touch.

5.13.6 Surface Preparation Requirements:

The chlorinated rubber undercoat will be fully cured, dry, clean, and free of oily or greasy contaminants.

5.13.7 Additional Remarks:

Coating systems other than chlorinated rubber can be used as the undercoating as long as they are' suitable for the service environment, are compatible with the non-skid paint, are fully cured, and have been spot tested to assure that the solvents in the non-skid paint are not detrimental to their adhesion.

5.14 Spar Varnish System

5.14.1 Type of Coating:

Phenolic resin modified oil varnish.

5.14.2 Use:

Primarily for exterior woodwork. Resistant to normal weathering, high humidity, and occasional wetting at ambient temperatures.

5.14.3 Reference Purchase Specification: 78-TMSS-04

5.14.4 Minimum Acceptable Dry Film Thickness (DFT): Not critical.

5.14.5 Approved Material:

Description	Application Method	Typical Recoating Interval(2)	Material Take-Off Guide (1).
			Typical Coverage sq.m/L
Spar varnish	Brush	24 Hours min.	12.3

Wood Stains For modifying wood color before varnish application.
Wood Filler Paste To fill large pores or openings.
Mineral Spirits Thinner and equipment cleaner.
Turpentine Solvent.

NOTE:

(1) For ordering purposes only. Not a mandatory painting specification.
(2) For guidance only. Recoat' when dry to touch.

5.14.6 Surface Preparation Requirements: Paragraph 6.2.8

5.14.7 Additional Remarks:

a. Shelf life: 12 months at 40°C.

b. If wood stain or filler paste has been used, allow them to dry hard before applying varnish.

c. On very absorptive wood, the first coat can be thinned with turpentine or mineral spirits.

d. For durability, three coats are recommended.

e. For maximum gloss, each dried coating should be lightly sanded with fine sand paper or steel wood prior to the application of the next coat.

5.15 Interior Lacquer System

5.15.1 Type of Coating:

Solution of nitrocellulose, resins, and plasticizers, with or without colored pigments.

5.15.2 Use:

Decorative for interior wood and metal under mild service conditions at normal ambient temperatures. Not resistant to exterior weathering or corrosive environments.

5.15.3 Reference Purchase Specifications: 78-TMSS-04

5.15.4 Minimum Acceptable Dry Film Thickness (DFT) Not critical.

5.15.5 Approved Material:

Description	Application Method	Typical Recoating Interval(2)	Material Take-Off Guide (1).
			Typical Coverage Sq.m/L
Sealer, Lacquer	Spray	2 hours	9.8
Lacquer	Spray	4 hours	12.3

NOTE:

(1) For ordering purposes only. Not a mandatory painting specification.
(2) For guidance only. Recoat when dry to touch.

5.15.6 Surface Preparation Requirements:

Wood, see Section 6.2.8
Metal, see Section 6.2.5

5.15.7 Additional Remarks:

a. Sand down the seal coat with fine sand-paper. On coarse grained woods, apply a second coat of sealer and repeat sanding.

b. Allow each coats to dry hard before applying the next coat. Typical application consists of one or two coats of sealer and two or more coats of lacquer.

c. Make sure the compressed air in the spray equipment is properly moisture-freed. Presence of moisture will spoil the final appearance by resulting in blushing. Do not apply in high humidity atmospheres.

5.16 Fluorescent Enamel System:

5.16.1 Type of Coating:

Alkyd resins containing fluorescent dyes.

5.16.2 Use:

For improved visibility at dawn or dusk, on metallic and non-metallic surfaces, do not use for traffic signs. Do not use on barricades or obstructions unless a reflective paint system is also provided.

5.16.3 Reference Purchase Specification:

Manufacturer technical brochure and specifications.

5.16.4 Minimum Acceptable Dry Film Thickness (DFT):

White Enamel Undercoat Not critical
Fluorescent Paint 0.080 mm
Protective Topcoat 0.040 mm

5.16.5 Approved Material:

Description	Application Method	Typical Recoating Interval(2)	Material Take-Off Guide (1).	
			Typical Coverage Sq.m/L	Typical Dft Per Coat mm
White Enamel	Refer Sections 5.7 and 5.8			
Fluorescent Paint	Brush or spray	1 hour	7.4	0.040
Protective Topcoat	Brush or Spray	N/A	6.1	0.040

Mineral Spirits Thinner and equipment cleaner.

NOTE: (1) For ordering purposes only. Not a mandatory painting specification.
(2) For guidance only. Recoat when dry to touch.

5.16.6 Surface Preparation Requirements:

White enamel undercoat must be dry and free of all surface contaminants.

5.16.7 Additional Remarks:

a. Composition of the fluorescent paints is critical. No pigments will be added. These will destroy the fluorescent capability of the paint.

b. All fluorescent paints deteriorate when exposed to sunshine. The proprietary topcoat retards but does not halt this deterioration. Typically, fluorescence is retained for about a year when the topcoat is used and only 4-6 months when the topcoat is omitted.

5.17 Interior Fire-Retardant Paint System

5.17.1 Type of Coating:

Generally, alkyd and chlorinated rubber resins pigmented for various colors, but water based systems are also found.

5.17.2 Use:

Interiors of buildings or other enclosures where fire resistance is desirable.

5.17.3 Reference Purchase Specification:

Manufacturer technical brochure and specifications.

5.17.4 Minimum Acceptable Dry Film Thickness (DFT):

Not critical.

5.17.5 Approved Material:

Description	Application Method	Typical Recoating Interval(2)	Material Take-Off Guide (1).
			Typical Coverage sq.m/L
Primer/Sealer	Brush, Spray or Roller	24 hrs	3.7
Topcoat	Brush, Spray or Roller	24 hours	8.6

Enamel Thinner Thinner and equipment cleaner.

NOTE:(1) For ordering purposes only. Not a mandatory painting specification.
(2) For guidance only. Recoat when dry to touch.

5.17.6 Surface Preparation Requirements:

a. Non-metallic surfaces – See Section 6.2.8

b. Metallic surfaces – As per Section 6.4.1 plus one coat zinc Phosphate primer.

c. Previously painted surfaces - dry and free of surface contaminants.

5.18 Inorganic Zinc Silicate Systems

5.18.1 Type of Coating:

Metallic zinc in either a water-base or ethyl-base silicate vehicle. Self-curing.

5.18.2 Use:

The required primer on new construction using the Section 5.1 epoxy/epoxy-Phenolic paint system. A substitute for the epoxy primer in Section 5.1 for maintenance painting subject to the conditions in Section 5.18.7 below. Can also be used as a substitute for the high temperature coating system primer in Section 5.12 up to a temperature of 400°C.

5.18.3 Reference Purchase Specification: 78-TMSS-03

5.18.4 Minimum/maximum Acceptable Dry Film Thickness (DFT):

0.060/0.100 mm

5.18.5 Approved Material:

Description	Application Method	Typical Recoating Interval	Curing Time	Material Take-Off Guide (2).	
				Typical Coverage sq.m/L	Typical Dft Per Coat mm
Zinc-silicate (water-based)	Spray	10 hrs	24 hrs(l)	7.4	0.080
Water	Thinner and equipment cleaner in water-based zinc-silicates				
Zinc-silicate (ethyl-based)	Spray	6 to 24 hrs(l)	24/48 hrs(l)	7.4	0.080

Ethanol Thinner and equipment cleaner for ethyl-based zinc-silicates.

NOTE:

(1) Varies considerably depending on manufacturer, temperature humidity, and wind. See Manufacturers' instructions. "Curing time" is the minimum allowable time interval between final painting and putting into service.

(2) For ordering purposes only. Not a mandatory painting specification.

5.18.6 Surface Preparation Requirements: As per Section 6.2.6.

5.18.7 Additional Remarks:

a. Ethyl-based zinc-silicates should be used when application is made under conditions of high humidity. The service performance of the fully cured ethyl-based or water based coatings are similar.

b. Surface preparation is critical.

c. Inorganic zinc silicate cannot be used over zinc-Phosphate.

d. Section 5.18 cannot be top coated by alkyd paint systems such as, but not necessarily restricted to Section 5.5, 5.6 and 5.7.

5.19 Traffic and Zone marking Paint Systems

5.19.1 Type of Coating:

Generally based on alkyd and chlorinated rubber resins. Lacquers and epoxies are also used.

5.19.2 Use:

Traffic and zone marking on concrete, asphaltic, or bituminous surfaces. Typical uses are for road lines, designating parking and no parking areas, and marking pedestrian crosswalks.

5.19.3 Reference Purchase Specification:

Manufacturer technical brochure and specifications.

5.19.4 Minimum Acceptable Dry Film Thickness (DFT): Not critical.

5.19.5 Approved Material:

Description	Application Method	Typical Drying Time Before Use	Material Take-Off Guide (1). Typical Coverage Sq.m/L
White (20 L) Yellow (5 L) Yellow (20 L)	Brush, Spray or Roller	1 hrs	4.9 TO 9.8(l)

Lacquer Thinner (2) Thinner and equipment cleaners.

Enamel Thinner (2)

Glass Beads For use when reflectivity is desired.

NOTE:

(1) Depends on thickness applied.
(2) Depends on particular system being used. Consult Manufacturers' instructions. Do not use aromatic thinners.
(3) For ordering purposes only. Not a mandatory paint specification.

5.19.6 Surface Preparation Requirements: As per Section 6.2.8.

5.19.7 Additional Remarks:

a. These are not multiple coat systems. The thickness required will usually depend on the roughness of the surface being coated, rougher surfaces requiring thicker coats.

b. If glass beads are used in added reflectivity, they should be sprinkled on right after the coating has been applied, using approximately 2.4 kg of beads per liter of paint. Paint systems containing glass beads premixed into the paint are acceptable as long as immediate reflectance is not required and the paint will be applied by brush or roller. (The beads are too large for spray application.)

c. Painting should not be done during foggy weather or in the presence of blowing sand.

5.20 Fiberglass Reinforced Epoxy Coatings

5.20.1 Type of Coating:

Epoxy resins reinforced with embedded fiberglass.

5.20.2 Use:

For coating -steel surfaces where additional strength, flexibility, corrosion protection and abrasion resistance of fiberglass - reinforced coating is desirable. Primary use is for repairs of tank bottoms, sides, and roofs that are severely corroded and require either large-scale repair or replacement.

5.20.3 Reference Purchase Specification:

78-TMSS-01, 78-TMSS-02, and manufacturer information

5.20.4 Minimum Acceptable Dry Film Thickness (DFT):

a. Hand lay-up 2.290 mm

b. Spray application 1.270 mm

5.20.5 Approved Material:

Description	Application Method	Min./Max Recoat Interval	Typical Curing Time Before Service	Typical Coverage Sq. m/L
Primer	Spray	4/24 hrs at 25 to 38^{0}C		0.080 to 0.130
(A) Hand lay-up with fiber- glass mats	Roller Hand		5 to 7 days (See 3.20.8.j)	2.290 to 2.790
(B) Spray applied system	Spray		5 to 7 days	1.270 to 1.520

5.20.6 Surface Preparation Requirements:

a. Hand Lay-up: Paragraph 6.2.6 and surface profile of 0.090 mm.

b. Spray Type: Paragraph 6.2.6 and surface profile of 0.050 mm.

5.20.7 Instructions:

a. For Strength and Corrosion Protection Hand Lay-up:

i. Coat with the approved primer 0.080 to 0.130 dry mm to - blasted surface before oxidation forms.

ii. Apply 0.510 to 0.640 mm of catalyzed, solvent free resin to primed surface.

iii. Apply one layer of 460 g/m^2 fiberglass mat to wet resin and smooth mat by hand to remove wrinkles, then, roll mat with steel rollers to remove trapped air and. insure good contact with the substrate and complete penetration of resin into the mat.

iv. Apply second coat of catalyzed resin at 0.640 to 0.760 mm to just rolled mat.

v. Apply 680 g woven roving to this second coat of resin and repeat smoothing and rolling technique as after first coat of resin.

vi. Apply 0.760 to 0.890 mm of catalyzed resin.

vii. Apply a layer of 45 g mat and do smoothing/rolling operation again.

viii. Finally, apply 0.250 mm of catalyzed resin to this mat and smooth out. Total dry film thickness shall be 2.290 to 2.790 mm.

b. For Corrosion Protection Spray Application:

i. Proceed to blast and prime as described in Method A and B. In the case of riveted tanks, make rivets flush with an epoxy caulking compound. In welded tanks, lap welds shall be made flush with epoxy caulk.

ii. Spray apply solvent free resin 45 g chopped fiberglass mix to the prepared surface. Roll the sprayed surface immediately to embed the fiberglass strands. The rolled coating thickness shall be a minimum of 1.020 mm.

iii. Within 24 hours, spray 0.250 to 0.510 mm of epoxy resin to layer of fiberglass coating to ensure all glass strands/fibers are covered. Total dry film thickness shall be 1.270 to 1.520 mm.

5.20.8 Additional Remarks:

a. Any necessary mechanical repairs and/or thorough cleaning of oil, grease or other contaminants shall be completed before abrasive blasting is started.

b. For tank bottoms, the minimum areas to be coated shall be entire bottom and 1 m up the sides.

c. Blast cleaning shall extend a minimum of 150 mm beyond the area to be coated to ensure good adhesion at the edges.

d. The blasted surface shall be cleaned of all debris by brushing or blowing with air followed by vacuum cleaning before being primed. NG inspector should examine the blast-cleaned area before priming begins to ensure that it meets standards.

e. If it is necessary to walk on the surface being coated (such as in the case of tank bottoms), rubber soled boots are the only allowable footwear once blasting and priming have begun. Access to a tank bottom shall be restricted to a single man way which shall have a 2.5m x 2.5m steel grating or plywood pad (fitted with a fiber foot scraper) placed immediately adjacent to it.

f. For floors being coated, once priming is completed, the floor shall be swept absolutely free of debris. Thereafter, the only equipment permitted on the floor are lights, fiberglass cloth, brush, rollers, and serrated rollers. Ideally, the resin should be mixed outside and passed in through the access man way.

g. After priming, all fillets (such as shell-to-wall bottom joints of tanks) shall be filled with an epoxy - sand (or equivalent) patching compound so that there are no sharp angles to be laminated. The minimum throat dimension of the shell-to-bottom fillet shall be 25 mm. Other fillets may be smaller.

h. Entrapped sand or other foreign matter in the coating is unacceptable. Entrapped air bubbles are unacceptable. Wrinkles, sags, dry spots and excessive thickness variations in the coating are unacceptable. All these defects shall be removed and the affected areas repaired.

i. The cured coating shall be 100% inspected with a holiday detector calibrated to detect a pinhole in a sample plate coated in the same way and having the same thickness as the surface in question. All these defects shall be removed and the affected areas repaired.

j. The coated surface shall be allowed to cure for the length of time recommended by the manufacturer (usually several days) before being placed in service. During this time, the surface shall be kept free of sand, dust, and moisture.

5.21 Mastic Coating for Immersion Service

5.21.1 Type of Coating:

Bituminous mastic.

5.21.2 Use:

For steel in immersion service up to 93°C where either the long term protection afforded by Section 5.3 is not needed or where the surface preparation required by Section 5.3 cannot be obtained. Suitable for drinking water tanks.

5.21.3 Reference Purchase Specification: 78-TMSS-02 and Manufacturer technical information.

5.21.4 Minimum Acceptable Dry Film Thickness (DFT): 0.510 mm

5.21.5 Approved Material:

Description	Application Method	Typical Recoating Interval(2)	Material Take-Off Guide (1).	
			Typical Coverage Sq.m/L	Typical Dft Per Coat
Mastic	Brush	24 hrs	2.4	0.250

Enamel Thinner Equipment cleaner.
Mineral Spirits Equipment cleaner.

NOTE:

(1) For ordering purposes only. Not a mandatory painting specification.
(2) For guidance only. Recoat when dry to touch.

5.21.6 Surface Preparation Requirements: as per Section 6.2.3

5.21.7 Additional Remarks:

Section 5.11 mastics may not be substituted for the Section 5.21 mastics in any service where the water might be used for drinking. Mastics approved for use under Section 5.21 shall be certified by a recognized institution as suitable for use in drinking water service. Paint Certificate shall bear approval by the Ministry of Health of Saudi Arabia.

5.22 Epoxy Coatings for Application on Damp Steel Surfaces

5.22.1 Type of Coating:

Catalytically cured epoxy coating systems.

5.22.2 Use:

Primarily intended for use on off-shore steel structures and platforms in above the water areas at temperatures up to 150°C. Typical exposures are sea water atmospheres with occasional wetting. This system may be applied to slightly damp surfaces and in high humidity weather conditions at 10°C and above.

5.22.3 Reference Purchase Specification: 78-TMSS-02

5.22.4 Minimum Acceptable Dry Film Thickness (DFT): Total = 0.380 mm

5.22.5 Approved Material

Description	Application Method	Min./Max Recoat Interval(1)	Cure Time(2)	Material Guide (3).	
				Typical Coverage Sq.m/L	Typical Dft Per Coat mm
Primer	Brush or Spray	2/24 hrs 4/24 hrs	5/3 days 5/3 days	11.0 4.4	0.050 0.130
Topcoat	Brush, or Spray	4/24 hrs 8/24 hrs	5/3 days 5/3 days	11.0 4.4	0.050 0.130

MEK Equipment cleaner and thinner (mix with 50%Toluene)
Toluene Equipment cleaner and thinner (must mix with 50% MEK)

NOTE:

(1) Min. /max. Recoat time at 25°C.
(2) Curing time for temperatures of 21°C/38°C.
(3) For ordering purposes only. Not a mandatory specification.

5.22.6 Surface Preparation Requirements:

See section 5.2.6. The minimum allowable anchor pattern (surface profile) is 0.060 mm.

5.22.7 Additional Remarks:

a. Shelf life: 12 months at 40°C.

b. Mix pigmented component, then add catalyst while stirring. The completely mixed paint should be allowed to stand 20 to 30 minutes before use.

c. Rollers will not be used. Preferred application method is by airless spray unit.

d. The catalyst is strongly alkaline and should be washed off immediately with water if splashed on the skin.

e. Welds, corners and edges should be brush painted prior to application of first spray coat to ensure adequate coverage in these areas.

f. The epoxy coatings of this system may not be applied on top of zinc Phosphate primer Section 5.5.

g. Follow manufacturer's instructions to determine type and amount of thinner to be used.

5.23 High Temperature Mastic Coatings

5.23.1 Type of Coating;

Cold Applied Black Mastic

5.23.2 Use:

Primarily for corrosion protection of buried steel pipe, tanks and other structures at service temperatures up to 93°C. Resistant to soil stresses, salts and moisture. Not suitable for exposure to sunlight.

5.23.3 Purchase Specification: 78-TMSS-02 and/or Manufacturer technical information.

5.23.4 Minimum Acceptable Dry Film Thickness (DFT): Total 0.510 mm

5.23.5 Approved Materials

Description	Application Method	Typical Recoating Intervals(2)	Material Take-Off Guide (1).	
			Typical Coverage Sq.m/L	Typical Dft Per Coat mm
Mastic	Brush, or Spray	24hrs	1.7	0.250

Xylene Equipment cleaner.

NOTE:

(1) For ordering purposes only. Not a mandatory painting specification.
(2) For guidance only. Recoat when dry to touch.

5.23.6 Surface Preparation requirements:

See section 6.2.5 (Commercial Blast)

5.23.7 Additional Remarks:

a. Shelf life: 12 Months at 40°C.

b. Materials are thixotropic. Do not add solvents to thin except when spraying at temperatures below 16°C. Stir thoroughly before using. When mixed vigorously, the mastics become free flowing.

c. Use Power stirrers. Hand stirring is totally inadequate.

d. Stir from time to time during application as required to maintain the desired viscosity.

e. When brushing mastic, it should be applied with a generous amount on the brush at all times.

f. Recommended spray equipment-Graco Bull dog or equivalent airless pump, 13 mm minimum material hose. For lengths over 7.6 m use 19 mm I.D. high pressure wire braid hose. Use mastic spray gun of good quality approved by NG Representative at site. Spray tip should be 0.9 mm minimum.

g. Maximum acceptable surface temperature for application varies with manufacturer.

h. Do not bury until the final coat of mastic is dry to touch.

6.0 SURFACE PREPARATION SPECIFICATIONS (SPS)

6.1 Acceptable Methods of Surface Preparation

6.1.1 Chemical and Solvent Wiping:

a. Solvent degreasing.
b. Alkali cleaning.
c. Steam cleaning.
d. Detergent washing.
e. Acid pickling or etching.

6.1.2 Surface Pretreatments:

a. Wetting oil pretreatment.
b. Phosphating pretreatment, including proprietary.
c. Wash primers.

6.1.3 Mechanical Methods:

a. Grinding and sanding.
b. Wire brushing (hand and powered).
c. Chipping (hand and pneumatic).
d. Abrasive blasting.

6.2 Definitions of the Various Grades of Surface Cleanliness as per The Society for Protective Coatings (hereinafter referred to as SSPC)

6.2.1 SSPC-SP1, -Solvent Cleaning:

Removal of oil, grease, dirt, soil, salts, and contaminants by cleaning with solvent, vapor, alkali, emulsion, steam, or detergent.

6.2.2 SSPC-SP2. "Hand Tool Cleaning":

Removal of loose rust, mill scale and paint by hand chipping, scraping, sanding, or wire brushing. The resulting surface has a faint metallic sheen. Its appearance corresponds to the Swedish Pictorial Standards designated St2.

6.2.3 SSPC-SP3, 'Power-Tool Cleaning":

Removal of loose rust, mill scale, and paint by power tool chipping, sanding, wire brushing, or grinding. The resulting surface has a pronounced metallic sheen. Its appearance corresponds to the Swedish Pictorial Standards designated St3.

6.2.4 Sa1, "Brush-off Blast"

A surface preparation where all oil, grease, dirt, rust scale, loose mill scale, loose rust, and loose paint or coatings are removed completely, but tight mill scale and tightly-adhered rust, paint, and coatings are permitted to remain provided that all mill scale and rust have been exposed to the abrasive blast pattern sufficiently to expose numerous flecks of the underlying metal fairly uniformly distributed over the entire surface.

6.2.5 Sa2, "Commercial Blast":

A surface preparation where at least 2/3 of each element of area is in the white metal condition. All oil, grease, dirt, rust scale, and foreign matter shall be completely removed from the surface and all rust, mill scale, and old paint shall be completely removed except for slight shadows, streaks, or discolorations caused by rust stain, mill scale oxides, or slight, tight residues of paint or coating that may remain. If the surface is pitted, slight residues of rust or paint may be found in the bottom of pits.

6.2.6 Sa2-1/2, "Near-white Metal Blast”:

A surface where at least 95% of each element of area is in the white metal condition. All oil, grease, dirt, mill scale, rust, corrosion products, oxides, paint, or other foreign matter shall be completely removed from the surface except for very light shadows, very slight streaks, or slight discolorations caused by rust stain, mill scale oxides, or slight, tight residues of paint or coating that may remain.

6.2.7 Sa3, "White Metal Blast":

A surface with a gray-white, uniform metallic color slightly roughened to form a suitable anchor pattern for coatings. The surface, when viewed without magnification, shall be free of all oil, grease, dirt, visible mill scale, rust corrosion products, oxides, paint or any other foreign matter.

6.2.8 Surface Cleanliness Requirements for Non-Metallic Surfaces":

Non-metallic surfaces such as (but not necessarily limited to) wood, concrete, plaster and tile shall be free of all oil, grease, dirt, soil, salts, loose paint, and other surface contaminants before being coated.

In addition, the surface shall be dry and the moisture content of the material shall be within the allowable limits of recognized good painting practice before being coated. Glazed surfaces of materials, such, as concrete or plaster, shall be roughened by acid etching or mechanical means prior to coating.

6.3 Definitions of Service Conditions

6.3.1 Mild Service:

Services characterized by the absence of such factors as chemical exposure, immersion, brine, condensation, high humidity, burial, corrosive environments, high temperatures, or abrasion. Examples of mild services include (but are not limited to) atmospheric exposure in low humidity (i.e. inland desert) areas, occasional splash and spillage, and the interiors of buildings.

6.3.2 Severe Service:

Services characterized by such factors as chemical exposure, immersion, brine, condensation, high humidity, burial, corrosive environments, high temperature, or abrasion. Examples of severe services include (but are not limited to) atmospheric exposure in high humidity (i.e. coastal areas), frequent splash and spillage, furnace stacks, off-shore structures, tank bottom interiors, and where industrial fumes are concentrated.

6.4 Company Surface Preparation Specifications (SPS)

The following specifications give the minimum acceptable states of surface preparation for the various types of coating systems used by NG. The required surface preparation varies with the material being coated and with the type of service (severe or mild).

6.4.1 Surface Preparation Specifications for Alkyd, Enamel, and Varnish Coatings:

a. Steel, mild service: Sa2 or SP-2 plus wetting oil

b. Steel, severe service: Sa2

c. Aluminum or Galvanized, mild and severe service: Phosphating pretreatment (alternate for aluminum, Sa1)

d. Non-metallic, mild and severe service: See section 6.2.8

6.4.2 Surface Preparation Specifications for Vinyl and Chlorinated Rubber Coatings:

a. Steel, mild service: Sa2

b. Steel; severe service: Sa2-1/2

c. Non-metallic; mild and severe services: See section 6.2.8

6.4.3 Surface Preparation Specifications for Catalyzed Coal Tar Epoxy Coatings:

a. Steel; mild service: Sa2

b. Steel; severe service: Sa2-1/2

c. Aluminum or galvanized, mild and severe services: Phosphating pretreatment (alternate for aluminum, Sal)

d. Non-metallic, mild and sever e service: See section 6.2.8

6.4.4 Surface Preparation Specifications for Catalyzed Epoxy or Epoxy-Phenolic Coatings:

a. Steel; atmospheric service: Sa2-1/2

b. Steel; immersion service: Sa3

c. Aluminum or galvanized, mild and severe services: Phosphating pretreatment (alternate for aluminum, Sal)

d. Non-metallic, mild and severe services: See section 6.2.8

6.4.5 Surface Preparation Specification for Zinc-silicate Coatings:

Steel, severe service: Sa2-1/2

6.4.6 Surface Preparation Specifications for High Temperature Silicone-type Coatings:

Steel, severe service: Sa3

6.4.7 Surface Preparation Specifications for Cold-applied Bitumastics:

a. Steel; severe service: SP-2

b. Steel; severe service: SP-3

c. Aluminum or galvanized; mild and severe services: Phosphating pretreatment (alternate for aluminum, Sal).

d. Non-metallic, mild and severe service: See section 6.2.8

6.4.8 Surface Preparation Specifications for Latex Emulsion:

Non-metallic, mild service See 6.2.8

7.0 STORAGE AND APPLICATION SPECIFICATIONS

7.1 General Storage Requirements

7.1.1 Paints, catalysts and thinners shall be stored in well ventilated or air-conditioned buildings free from excessive heat, sparks, flames and direct sunlight. Paints stored for short times on the job site shall be protected from direct sunlight during the summer months.

7.1.2 Containers of paint, catalysts, and thinners shall not be opened prior to use except for spot inspection. They shall be tightly resealed after inspection.

7.1.3 New stocks or shipments of paint shall be stored in such a manner that older stocks are issued first. Paints that have exceeded their shelf lives as given in the manufacturer’s instructions shall not be issued or used without the approval of NG Representative.

7.1.4 Catalytically-cured protective coating "Kits" will not be issued or used if any of the cans of Components shows signs of leakage. All such instances shall be brought to the attention of NG Representative.

7.1.5 Paint storage area shall be provided with fire protection system (Detection and Suppression).

7.1.6 The disposal of residual or paints with expired validity shall be as per PME Standards.

7.2 General Mixing Requirements

7.2.1 Manufacturers' instructions and the requirements of the applicable NG Paint Standard Specification shall be followed.

7.2.2 The largest quantity of paint that can be mixed by hand is 20 liters. Larger amounts shall be mixed using power stirrers.

7.2.3 Mixing shall be done in a well-ventilated area free of sparks and open flames.

7.2.4 If a skin has formed over the surface of the paint, it shall be carefully cut loose and lifted out of the can prior to mixing.

7.2.5 If sludge has formed in the bottom of the paint can due to settling, mixing shall be continued until the paint is smooth and homogeneous. If the sludge cannot be re-dispersed by mixing, the paint shall not be used.

7.2.6 Paint that appears gritty or contains foreign matter shall be filtered through cheesecloth (or equivalent) before use.

7.2.7 When coating systems that settle quickly are being applied by spray, the spray pots shall be equipped with agitators to ensure continued mixing.

7.3 General Thinning Conditions

7.3.1 Manufacturers' instructions and the requirements of the applicable NG Paint Standard Specifications shall be followed.

7. 3.2 Excessive thinning of inherently viscous coating systems to improve ease of application shall not be allowed due to detrimental effects on the coating I s service performance.

7.3.3 Thinning shall be supervised by competent personnel who are acquainted with the correct amounts and types of thinners for the paint system being applied. Indiscriminate thinning by individual painters shall not be allowed.

7.4 General Application Requirements

7.4.1 No coating system shall be applied to surfaces that are structurally or superficially damp with the exception of coating systems specifically designed for those conditions.

7.4.2 Regardless of any additional requirements for a particular coating system, no Coating shall be applied over surfaces containing oil, grease, dirt, loose paint, loose rust, or other surface contaminants. Weld splatter and other sharp surface protrusions shall be flush removed from the surface prior to painting.

7.4.3 No coating system shall be applied to any surface whose temperature is below 5°C or above 60°C unless the coating is especially formulated for these conditions.

7.4.4 Usage of thinners on different ambient conditions shall be within the recommendation of the thinner manufacturer.

7.4.5 On exceptionally rough, pitted, or poorly cleaned surfaces, the primer coat shall be brushed on.

7.4.6 Weld seams, fillets, edges, bolts, rivets and other similar features shall be given an extra coat (so-called "Stripe Coat") of paint. Preferably, this coat shall be applied by brush before the normal coating sequence is started.

7.4.7 When coating is to be done for corrosion protection, alternate, or "signal", coats shall be of different colors to aid in obtaining complete coverage.

7.4.8 No coatings shall be applied under adverse weather conditions such as, but not restricted to, the situations below. The NG Representative’s decision will be final in this regard.

a. High Winds: Winds strong enough to blow sand, salt spray, or other foreign matter onto the surface being painted. Also, winds strong enough to cause "dry spray" or to seriously disturb the spray pattern when application is by spray gun.

b. High Humidity: If the streaks left by wiping a moist rag across the surface to be painted do not completely disappear within 5 minutes, a condition of "high humidity" exists. Only coatings specifically designated for application on a damp or wet surface may be applied while this condition persists. Do not use salt water to wet the rag for this test.

c. Excessively low or high temperatures; see section 7.4.3.

7.4.9 The finished coating shall be generally free of runs, sags, ridges, drips, and embedded foreign matter. Areas that are blistered, cracked, porous, uncoated or below minimum acceptable thickness shall be repaired.

7.4.10 For all internal coating systems in immersion service, desert sand is not allowed for use as the blasting medium.

7.4.11 Structural steel or equipment purchased for shipment to National Grid SA shall be fully coated with primer and 0.010 mm minimum DFT aluminum pigmented alkyd to avoid rusting during transit or construction phase. Primer alone is not acceptable. The aluminum transit barrier coating should then be primed with 0.020 mm DFT of zinc Phosphate and appropriate topcoat.

7.5 Specific Requirements for Roller Application

7.5.1 Rollers shall not be used when close control over dry film thickness is required.

7.5.2 Painting with long nap rollers is desirable on open work such as chain links, fences, etc.

7.5.3 Rollers shall be cleaned after usage.

7.6 Specific Requirements for Brush Application

7.6.1 The maximum brush size for all critical work shall be 125 mm. The bristles on 100 mm and 125 mm brushes will be approximately 100 mm long.

7.6.2 Brushes shall be cleaned after usage.

7.7 Specific Requirements for Spray Application

7.7.1 Coating systems to be sprayed should be filtered through layered cheese-cloth or equivalent to remove lumps and foreign matter before being put into the spray pot.

7.7.2 The spray gun shall be held perpendicular to the surface being coated.

7.7.3 The motion of the spray gun shall be parallel to the surface being coated.

7.7.4 The spray gun shall be held about 150-230 mm from the surface being coated. The paint must be deposited wet.

7.7.5 Spraying shall be done in areas free of sparks, flames, unusually hot surfaces, strong winds, and operating electrical equipment that is not explosion proof.

7.7.6 When spraying is done in enclosed areas, the ventilation requirements of "Safety Instructions" shall apply.

7.7.7 Caustic solutions shall not be used for cleaning the spray gun. The spray gun shall not be immersed in solvents for cleaning.

7.8 Coating Compatibility Chart

7.8.1 Purpose:

This chart indicates the compatibility of many of the different possible combinations of paint systems used by National Grid SA. Its function is to indicate which combinations of top coat and base (or existing) coat are compatible and which ones are completely unsuitable.

7.8.2 Use:

Given below is a reference chart. Designers, users, inspectors, etc. may use this chart as a quick reference for determining if a proposed paint system is technically possible. Many of those marked "Yes" are technically possible but rarely, if ever, used. Combinations marked "No" are physically incompatible and may not be used.

Chart for Topcoat/Base Coat Compatibility

COATING BEING APPLIED

EXISTING COATING	ALKYD	LACQUER	EPOXY/ COALTAR EPOXY	SILICONE ALKYD	CHLORINATED RUBBER	LATEX EMULSION	INORGANIC ZINC
ALKYD	Yes	No	No	N/A	No	Yes(1)	No
LACQUER	Yes	Yes	No	N/A	No	Yes	No
EPOXY/ COAL TAR EPOXY	Yes(1)	Yes	Yes(2)	N/A	Yes(1)	Yes(1)	N/A
SILICONE ALKYD	Yes	No	No	Yes	No	Yes	No
CHLORIN-ATED RUBBER	Yes	No	No	N/A	Yes	Yes	No
LATEX EMULSION	Yes	Yes	No	N/A	Yes	Yes	No
INORGANIC ZINC	No	No	Yes	Yes	Yes	Yes	Yes(3)

NOTES:

(1) Gloss must be removed from existing coating by weathering or sanding before applying topcoat.
(2) Topcoat must be applied before existing coating has cured.
(3) Only in the wet and after approval from National Grid Representative.

8.0 SAFETY

8.1 Fire and Explosion Hazards from Solvents

8.1.1 Flammable Materials:

In paint systems, it is normally the solvent vapor that is flammable and generally the other components are less dangerous and non-explosive. In the National Grid SA paint system, all the solvents except water present a fire explosion hazard.

8.1.2 Flash Point - Definition:

The flash point is defined as the lowest temperature at which a liquid will give off sufficient vapor to ignite when exposed to an open flame. For most paint solvents in the National Grid SA system, the flash point is less than the normal ambient temperatures in Saudi Arabia. The danger of fire exists virtually always where solvents are in use.

8.1.3 Flammable (Explosive) Limits-Definition:

The lower and upper flammable (explosive) limits define the range of vapor/air ratios that are potentially explosive. The lower flammable (explosive) limit (LEL) can be as low as 1% by volume - a level readily obtained in the area near opened solvent containers and near the nozzle of a spray painting gun in operation.

8.1.4 Ventilation:

a. Ventilation is necessary because nearly all solvent vapors are heavier than air. Therefore, they tend to settle to the lower level of confined areas.

b. Natural ventilation through opened man ways, etc. is rarely, if ever, sufficient to keep local vapor concentrations to a safe level. As a general rule, forced ventilation shall be used, especially in small enclosures, and always during spray painting.

c. Ventilation requirements are proportionately greater for vessels smaller than 10,000 barrels (bbls) capacity 1600 m^3 than for larger vessels. The ventilation requirements for various vessel sizes are shown in Table 8-1.

d. Mechanical or forced ventilation is needed during all spray painting operations in enclosed areas, whenever opened solvent containers are located in enclosed areas, in small enclosures during hand painting or solvent wiping, and whenever solvent vapors are irritating to eyes, skins, lungs, etc. Forced ventilation shall operate during paint application and continue until the coating is dry to the touch.

e. The fresh air inlet should be near the top of the confined space. The discharge openings should be located near the bottom and positioned to eliminate dead spaces. Supplementary fans may be needed to ensure good air circulation with no dead air pockets in the enclosed area.

f. Even with forced ventilation in the amounts given in Table 8-1, vapor concentrations during spray painting will be high. While spray painting is in progress, all workmen shall wear chemical cartridge respirators if outdoors or airline hoods if inside a vessels.

8.1.5 Fire Precautions:

a. Smoking or the use of open flames is permitted only in designated safe areas and never inside vessels.

b. All electrical lighting and equipment shall be explosion proof.

c. Solvents and paints shall not be applied to surfaces warmer than summer ambient temperatures.

d. Ventilating equipment shall be used for work in confined areas.

8.2 Health Hazards

8.2.1 General Note:

Preventive Medicine Services shall be advised prior to repainting of drinking water storage tanks to carry out contamination prevention and disinfection of the water system.

8.2.2 Toxicity:

a. Most of the solvents contained in paints or used for cleaning are toxic in varying degrees. The dangers can arise from inhalation, ingestion, or skin absorption.

b. Examples of acceptable solvents are mineral spirits and naphtha, enamel thinner, toluene, Xylene, methylethylketone, turpentine, and methylisobutylketone. These solvents are listed in order of increasing toxicity.

c. Examples of extremely toxic solvents are benzene, methyl cellulose, methyl cellulose acetate, and all chlorinated solvents such-as carbon tetrachloride. Because of their high toxicity, none of these solvents are used in National Grid SA paint systems.

d. The vapors from many solvents can cause mild to quite severe allergic skin irritations. The alkaline catalysts used in many epoxy paints can cause severe chemical burns if not washed off immediately with water.

8.2.3 Personal Protective Gear:

The use of the appropriate personal protective gear shown in Table 8-2 will greatly reduce most health problems.

8.3 Equipment Hazards

Much of the equipment used in surface preparation and paint application is hazardous. This section outlines the most common of these hazards.

8.3.1 Hazards from Hand Operated and Power Tools:

a. The principal hazards associated with hand power tools are flying particles and dust. The applicable personal protective gear shown in Table 8-2 shall be worn. Examples of power tools include pneumatic chippers and needle hammers, rotary wire brushes, and abrasive wheels. Examples of hand tools are chipping hammers, chisels, scrapers, and wire brushes.

b. Electrically driven power tools shall be properly grounded to prevent shock.

c. Power equipment shall be operated at the speeds recommended by the manufacturer and shall have proper safety guards.

8.3.2 Hazards from Sandblasting:

a. Dust from sandblasting operations constitutes a serious health hazard, both to the operator and to other workmen in the immediate area. The applicable personal protective gear listed in Table 8-2 shall be worn.

b. Sandblast nozzles shall be equipped with a remote control shutoff of the "deadman" variety to reduce the hazard of injury to nearby workmen if the hose is dropped.

c. The blast nozzle shall be electrically grounded to prevent static electrical discharges or shocks to operating personnel.

d. A noise hazard exists with sandblasting. Ear protection must always be worn when blasting particularly within confined areas. Other workmen in the immediate area may also need ear protection.

8.3.3 Hazards from Flame Cleaning:

A major danger of explosion and fire exists when flame cleaning is done in the presence of flammable materials such as paint solvents. Therefore, flame cleaning is not allowed.

8.3.4 Hazards from Solvent Cleaning:

a. Permissible solvents for wiping include mineral spirits, Stoddard solvent, petroleum naphtha, turpentine, Xylene, and toluene. Benzene, gasoline, carbon tetrachloride, and chlorinated hydrocarbons are dangerous and must not be used.

b. A fire and explosion hazard always exists when using solvents, especially in confined areas. Adequate ventilation must be supplied.

8.3.5 Hazards from Paint Preparation and Equipment Cleaning:

a. The solvents for all National Grid SA paint systems, except water based paints, are toxic and flammable. These paints should be mixed either outside or in an adequately ventilated area.

b. Adequate washing facilities should be readily available so that paints and solvents splashed on the body or in the eyes can be immediately removed.

Electrically driven power mixers shall be grounded.

8.3.6 Hazards from Paint Application:

a. Spray paint results in local high solvent vapor concentrations, giving rise to serious fire and explosion hazards. Spray painting must never be done either on or near surfaces hotter than summer ambient temperatures or near open flames. Adequate ventilation must be maintained in confined areas. See Table 8-1.

b. Airless spray guns operate at very high pressures in the range of 13.8 to 17.2 MPa. They are extremely hazardous; the paint jet, which is strong enough to easily slice through a man, at distances up to about 300 mm from the gun nozzle, is invisible. Operators and adjacent workmen must be thoroughly aware of its potential destructive power. When the gun is not in use, the safety catch must be put on so that the trigger cannot be pulled accidentally.

TABLE 8-1

Recommended air flow to maintain solvent vapor concentration below 10% of the lower explosive limit in enclosed areas:

Tank Volume		Air Mover Capacity
Barrels (bbls)	m^3	l/s
100	16	472
1000	160	1180
5000	800	2832
10000	1600	4720

NOTE: Ventilating to 10% of the lower flammable limit reduces the likelihood of fire or explosion, but the toxic vapor content will be too high to breathe. Supplied air respiratory protection is required inside confined spaces.

TABLE 8-2A Personal Equipment And Other Equipment Required To Be Either Worn Or Used During Surface Preparation And Paint Application (1)

Type Of Work	Hood Airline (2)		Respirator				Face Shield (3)		Goggles, Safety Impact	
			Chemical Cartridge		Type H Filter					
Surface Preparation	O	C	O	C	O	C	O	C	O	C
Wire Brushing, chipping, Scraping and grinding					X	X	X	X	X	X
Sand blasting -Operator -Other Workmen in the vicinity	 X X	 X X			X		X		X	
Paint Removing				X			X			
Solvent cleaning				X			X			

Paint Application	Brush		Spray		Brush		Spray		Brush		Spray	
	O	C	O	C	O	C	O	C	O	C	O	C
Hot applied Coal tar and asphalt coatings						X			X	X		
Cold applied coal tar coatings						X						
Coal tar epoxy coating				X		X			X	X	X	
Alkyd Type coating						X		X				
Phenolic-varnish coating						X		X				
Catalytically cured Phenolic and Phenolic epoxy coatings				X		X			X	X	X	
Vinyl Coating	X			X	X		X		X		X	
Wash Primer					X							
Inorganic Zinc					X					X		
Chlorinated Rubber Paints						X		X				
Lacquers		X		X	X		X		X		X	
Water base Paints												
Shellac						X			X		X	

TABLE -8-2B Personal Equipment And Other Equipment Required To Be Either Worn Or Used During Surface Preparation And Paint Application (1)

Type Of Work	Cover-Alls (6)	Gloves; Leather (6)	Gloves Rubber (6)	Cream Protective Ply 1,2,3	Ear Protection (6)	Washing Facilities	Air Movers (4)
Surface Preparation							
Wire Brushing, chipping, Scraping	X	X			C		
Sand blasting -Operator -Other Workmen in the vicinity	X X	X			C		
Paint Removing	X	X	X	X		X	X
Solvent cleaning	X		X	X		X	X

Type Of Work	Cover-alls (6)	Gloves (6)		Cream Protective Ply			Washing Facility	Air Movers (4)
Paint Application		Leather	Rubber	1	2	3		
Hot applied Coal tar and asphalt coatings	X	X			X		X	
Cold applied coal tar coatings	X	X			X		X	X
Coal tar epoxy coating	X	X			X		X	X
Alkyd Type coating	X	X		X			X	X
Phenolic-varnish coating	X	X		X			X	X
Catalytically cured Phenolic and Phenolic epoxy coatings	X	X			X		X	X
Vinyl Coating	X	X			X		X	X
Wash Primer	X	X					X	X
Inorganic Zinc	X	X				X	X	X
Chlorinated Rubber Paints	X	X			X		X	X
Lacquers	X	X		X			X	X
Water base Paints	X	X				X	X	X
Shellac	X	X				X	X	X

NOTES: O=Outdoor C=Confined Areas

(1) Coveralls and safety hats or airline hoods shall always be used; safety belts and lines shall be used by workman operating from unguarded platforms or in enclosed spaces where a manhole is the only exit.

(2) Airline hoods shall always be used in confined spaces having a volume of less than 19,000 liters, and as indicated.

(3) Face shields shall always be used when working overhead, and as indicated.

(4) Air movers shall be used in confined spaces as indicated, except where known sufficient ventilation is provided through open windows and doors.

(5) Some power cleaning tools such as chipping guns and pneumatic hammers are extremely noisy; ear protection shall be worn at all times.

(6) Various sizes and brands are available. Consult National Grid SA Stock Catalogue for details.